

# OpeX Industries — Centrifugal Pump CP-4400

## Installation, Operation and Maintenance Manual

The CP-4400 is a horizontal, single-stage, end-suction centrifugal pump designed for continuous industrial service. Rated flow is 440 cubic metres per hour at a total dynamic head of 62 metres. The pump casing is cast in duplex stainless steel (grade CD4MCu) for corrosion resistance in process-water applications. Maximum allowable working pressure is 16 bar at 120 degrees Celsius.

The standard driver is a 132 kW, 4-pole, 415V three-phase induction motor coupled via a flexible disc coupling. Nominal running speed is 1485 RPM. The mechanical seal is a balanced, single cartridge type rated to 25 bar. Bearing lubrication is grease-packed, relubrication interval every 4000 operating hours.

### Section 4: Casing Bolt Torque Specification

Casing bolts must be tightened in the sequence shown below, in three equal passes, to reach the final torque value. Use a calibrated torque wrench. Do not exceed the listed torque — overtightening can distort the casing split-ring seal face.

| Bolt position            | Bolt size | Torque (Nm) | Tightening sequence |
|--------------------------|-----------|-------------|---------------------|
| Suction flange, top      | M20       | 285         | 1                   |
| Suction flange, bottom   | M20       | 285         | 2                   |
| Discharge flange, top    | M24       | 460         | 3                   |
| Discharge flange, bottom | M24       | 460         | 4                   |
| Casing split, upper      | M16       | 165         | 5                   |
| Casing split, lower      | M16       | 165         | 6                   |
| Bearing housing          | M12       | 78          | 7                   |

### Section 5: Coupling and Baseplate Bolt Torque

The flexible coupling hub bolts and baseplate anchor bolts have separate torque requirements from the casing bolts. Recheck coupling bolt torque after the first 200 hours of running.

| Component                | Bolt size | Torque (Nm) | Recheck interval |
|--------------------------|-----------|-------------|------------------|
| Coupling hub, motor side | M10       | 48          | 200 hours        |
| Coupling hub, pump side  | M10       | 48          | 200 hours        |
| Baseplate anchor bolts   | M20       | 310         | 500 hours        |
| Motor foot bolts         | M16       | 175         | 500 hours        |

## **Section 6: Commissioning Checklist**

Before first start, confirm: (1) suction strainer is installed and clean, (2) casing is fully primed with no trapped air, (3) direction of rotation matches the arrow cast into the casing, (4) coupling guard is fitted, (5) baseplate grout has cured for at least 72 hours. Minimum continuous flow to prevent overheating is 65 cubic metres per hour. Do not run the pump dry for more than 30 seconds under any circumstances — dry running above this limit voids the mechanical seal warranty.

Vibration limits: normal operation should read below 4.5 mm/s RMS velocity at the bearing housings. Sustained readings above 7.1 mm/s require an immediate shutdown and inspection for bearing wear or cavitation.